

Algebraic Number Theory NMAL431

Homework 1

Due date: 03 November 2026 at 17:20

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1. Recall from lectures that a Dedekind domain is a domain which is Noetherian, integrally closed, and for which every non-zero prime ideal is maximal (i.e. the domain has Krull dimension ≤ 1).
 - (a) Find a domain which is Noetherian and integrally closed, but which has a non-zero prime ideal which is *not* maximal.
 - (b) Find a domain which is integrally closed and for which every non-zero prime ideal is maximal, but which is *not* Noetherian.
 - (c) Find a domain which is Noetherian and for which every non-zero prime ideal is maximal, but which is *not* integrally closed.

In each case, justify that your example has the required properties.

2. Let k be a field, and let $K := k(t)$ be the field of rational functions in one variable over k , i.e. $k(t) = \text{Frac}(k[t])$.
 - (a) Let $f \in k[t]$ be monic irreducible. Recall that every element $\alpha \in K^\times$ can be written as $\alpha = f^n g/h$ with $n \in \mathbb{Z}$ and $g, h \in k[t]$ not divisible by f . Define the function $v_f : K^\times \rightarrow \mathbb{Z}$ by $v_f(\alpha) := n$. Show that v_f is a valuation on K .
 - (b) Define the function $v_\infty : K^\times \rightarrow \mathbb{Z}$ by $v_\infty(g/h) := \deg h - \deg g$. Show that v_∞ is a valuation on K .
 - (c) Let $v : K^\times \rightarrow \mathbb{Z}$ be a valuation on K such that $v(k^\times) = 0$, i.e. v is trivial on $k^\times$. Show that either $v = v_f$ for some monic irreducible f , or $v = v_\infty$.

3. For each rational prime p , determine all elements $x \in \mathbb{Q}_p$ such that the series

$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

converges in $\mathbb{Q}_p$. (Here, convergence is with respect to the metric $d(x, y)$ given in lectures).